

Common Network Attacks — Guided Notes ANSWER KEY

Unit 4.1.2 | CompTIA Network+ N10-009 Obj. 4.1 | N10-008 4.2

For instructor use / distribute after students complete the notes.

Question 1

In a man-in-the-middle (MITM) attack, the attacker positions themselves _____ two communicating parties without either knowing. The attacker can both _____ and _____ the traffic.

Answer: Between; intercept (read); modify (alter or inject).

Question 2

ARP poisoning enables MITM attacks by sending _____ ARP replies that map the attacker's MAC address to a legitimate IP (usually the _____). This causes victim traffic to route through the attacker.

Answer: Gratuitous (unsolicited); default gateway.

Question 3

A Distributed Denial of Service (DDoS) attack differs from a DoS attack because it uses _____ sources, often a network of compromised devices called a _____. Why is this harder to block?

Answer: Multiple (distributed) sources; botnet. Blocking a single IP is ineffective when thousands of different IPs are sending traffic.

Question 4

A SYN flood exploits the TCP _____ handshake by sending a large number of SYN packets with _____ source IPs, causing the server to hold open half-connections until its table is exhausted.

Answer: Three-way; spoofed (forged).

Question 5

Explain how an evil twin attack works. What makes it more convincing than a simple rogue AP with a different SSID?

Answer: The attacker creates an AP with the same SSID (and often same BSSID/channel) as a legitimate AP. It's convincing because the network name is identical — clients may auto-connect or be deauthenticated from the real AP and forced to the fake one.

Question 6

DNS poisoning (cache poisoning) injects _____ records into a DNS resolver's cache, redirecting users to attacker-controlled IP addresses without the user changing any settings. The defense is _____.

Answer: Forged/malicious DNS; DNSSEC (DNS Security Extensions), which cryptographically signs records so forged responses are rejected.

Question 7

VLAN hopping using double tagging works when the attacker's port is in the same VLAN as the switch's _____. The attacker sends a frame with two 802.1Q tags — the outer tag is stripped at the first switch, exposing the inner tag and sending the frame to a _____.

Answer: Native VLAN; different (target) VLAN.

Question 8 — Real World Application

REAL WORLD: Users report their browser shows a certificate warning when loading the school portal, but only from the library Wi-Fi. No other network has this problem. Name the most likely attack and describe the two-step process an attacker used to create this symptom.

Answer: Evil twin / SSL stripping or MITM with certificate substitution. Step 1: attacker sets up a rogue AP matching the school SSID in the library. Step 2: clients connect to the rogue AP; the attacker intercepts HTTPS traffic and presents their own forged certificate, causing the browser warning.